

Name:

Date:

Solve each problem. Show your work in the box.

Write your final answer on the answer line (or in the blanks shown).

Use the strategies you learned in Lesson 5.11.08.

1 Draw a Triangle from Angles

You are given two angles of a triangle: 50° and 70° .

Find the third angle, then describe the steps to draw this triangle using a ruler and protractor.

Third angle = $180^\circ - 50^\circ - 70^\circ = 60^\circ$. Draw one side; mark 50° at one end and 70° at the other; extend the rays until they meet.

2 Draw a Quadrilateral

You need to draw a parallelogram with sides 6 cm and 4 cm and one angle of 70° .

List the steps and draw the shape.

Draw one 6 cm base; at one end draw a 70° angle; mark 4 cm along that ray; complete the parallelogram. Opposite angle = 70° .

3 Classify the Drawn Shape

A student draws a quadrilateral with all sides 5 cm and one angle measured as 90° .

What quadrilateral did she draw? Justify using properties.

A square — all four sides equal and one right angle means all four angles are 90° .

4 Triangle Impossible?

Kai tries to draw a triangle with sides 3 cm, 4 cm, and 9 cm.

Explain why this triangle cannot be drawn.

$3 + 4 = 7 < 9$. The sum of the two shorter sides must exceed the longest side. This triangle is impossible.

5 Reasoning: Design a Flag

Leo is designing a flag patch shaped like an equilateral triangle with each side 8 cm.

What are the measures of all three angles? If he cuts the triangle in half vertically, what are the angles of each resulting triangle?

All angles = 60° . Cutting vertically creates two right triangles each with angles 90° , 60° , and 30° .